CASE STUDY 1: PROGRESSIVE ELECTRICITY BILLING

1. Problem Statement

The aim of this case study is to develop a Python program to calculate a household electricity bill based on the number of units consumed. The program uses progressive electricity charges, fixed charges, and a rebate for low electricity consumption. It displays the final bill amount.

2. Objectives

1. To understand variables, data types, and arithmetic operators in Python.
2. To use conditional statements to calculate electricity charges.
3. To apply progressive billing slabs.
4. To calculate fixed charges and rebates.
5. To display the final electricity bill in a clear format.

3. Assumptions

1. Electricity units are entered by the user.
2. Units consumed cannot be negative.
3. The first 100 units cost Rs. 2 per unit.
4. The next 100 units cost Rs. 3 per unit.
5. Units above 200 cost Rs. 5 per unit.
6. A fixed charge of Rs. 50 is added to the bill.
7. A rebate of Rs. 20 is provided if consumption is 100 units or less.
8. These rates are assumed for this academic example.

4. Algorithm

1. Start the program.
2. Read the number of electricity units consumed.
3. Check whether the entered units are negative.
4. If the units are negative, display "Invalid units".
5. If units are less than or equal to 100, calculate the charge at Rs. 2 per unit.
6. If units are between 101 and 200, charge Rs. 2 for the first 100 units and Rs. 3 for the remaining units.
7. If units are above 200, charge Rs. 2 for the first 100 units, Rs. 3 for the next 100 units, and Rs. 5 for the remaining units.
8. Add the fixed charge of Rs. 50.

9. If consumption is 100 units or less, subtract the rebate of Rs. 20.
10. Display the energy charge, fixed charge, rebate, and total bill.
11. Stop the program.

5. Flowchart

START

↓

Input electricity units

↓

Are units less than 0?

- YES → Display "Invalid units" → END
- NO → Continue

↓

Are units less than or equal to 100?

- YES → Energy Charge = Units × 2
- NO → Check the next condition

↓

Are units less than or equal to 200?

- YES → Energy Charge = 100 × 2 + (Units − 100) × 3
- NO → Energy Charge = 100 × 2 + 100 × 3 + (Units − 200) × 5

↓

Add fixed charge of Rs. 50

↓

Are units less than or equal to 100?

- YES → Apply rebate of Rs. 20
- NO → Rebate = Rs. 0

↓

Calculate total bill

↓

Display bill details

↓

END

Note: Draw this flowchart using standard flowchart symbols in your assignment.

6. Pseudocode

```
BEGIN
INPUT units
IF units < 0 THEN
PRINT "Invalid units"
ELSE
IF units <= 100 THEN
    energy_charge = units * 2
ELSE IF units <= 200 THEN
    energy_charge = 100 * 2 + (units - 100) * 3
ELSE
    energy_charge = 100 * 2 + 100 * 3 + (units - 200) * 5
END IF
fixed_charge = 50
IF units <= 100 THEN
    rebate = 20
ELSE
    rebate = 0
END IF
total_bill = energy_charge + fixed_charge - rebate
PRINT energy_charge
```

```
PRINT fixed_charge

PRINT rebate

PRINT total_bill

END IF

END
```

7. Python Source Code

```
units = int(input("Enter electricity units consumed: "))

if units < 0:
    print("Invalid units")
else:
    if units <= 100:
        energy_charge = units * 2
    elif units <= 200:
        energy_charge = 100 * 2 + (units - 100) * 3
    else:
        energy_charge = 100 * 2 + 100 * 3 + (units - 200) * 5

    fixed_charge = 50

    if units <= 100:
        rebate = 20
    else:
        rebate = 0

    total_bill = energy_charge + fixed_charge - rebate

    print("\n--- Electricity Bill ---")
    print("Units consumed:", units)
    print("Energy charge: Rs.", energy_charge)
    print("Fixed charge: Rs.", fixed_charge)
    print("Rebate: Rs.", rebate)
    print("Total bill: Rs.", total_bill)
```

8. Test Cases

Test Case 1: Low Electricity Consumption

Input:

80

Calculation:

Energy charge = 80 × 2 = Rs. 160

Fixed charge = Rs. 50

Rebate = Rs. 20

Total bill = 160 + 50 − 20 = Rs. 190

Expected Output:

Enter electricity units consumed: 80

--- Electricity Bill ---
Units consumed: 80
Energy charge: Rs. 160
Fixed charge: Rs. 50
Rebate: Rs. 20
Total bill: Rs. 190

Test Case 2: Medium Electricity Consumption

Input:

150

Calculation:

First 100 units = 100 × 2 = Rs. 200

Remaining 50 units = 50 × 3 = Rs. 150

Energy charge = Rs. 350

Fixed charge = Rs. 50

Rebate = Rs. 0

Total bill = 350 + 50 = Rs. 400

Expected Output:

Enter electricity units consumed: 150

--- Electricity Bill ---
Units consumed: 150
Energy charge: Rs. 350
Fixed charge: Rs. 50
Rebate: Rs. 0

Total bill: Rs. 400

Test Case 3: High Electricity Consumption

Input:

250

Calculation:

First 100 units = 100 × 2 = Rs. 200

Next 100 units = 100 × 3 = Rs. 300

Remaining 50 units = 50 × 5 = Rs. 250

Energy charge = Rs. 750

Fixed charge = Rs. 50

Rebate = Rs. 0

Total bill = 750 + 50 = Rs. 800

Expected Output:

Enter electricity units consumed: 250

--- Electricity Bill ---
Units consumed: 250
Energy charge: Rs. 750
Fixed charge: Rs. 50
Rebate: Rs. 0
Total bill: Rs. 800

Test Case 4: Invalid Electricity Consumption

Input:

-5

Expected Output:

Enter electricity units consumed: -5
Invalid units

9. Sample Output

Enter electricity units consumed: 80

--- Electricity Bill ---
Units consumed: 80
Energy charge: Rs. 160
Fixed charge: Rs. 50
Rebate: Rs. 20
Total bill: Rs. 190

10. Conclusion

This case study demonstrates how Python can be used to calculate electricity bills using progressive billing slabs, fixed charges, and rebates. It uses variables, arithmetic operators, conditional statements, and input/output functions. The program also validates negative input values and displays the final electricity bill clearly.

```
units = int(input("Enter electricity units consumed: "))

if units < 0:
    print("Invalid units")
else:
    if units <= 100:
        energy_charge = units * 2
    elif units <= 200:
        energy_charge = 100 * 2 + (units - 100) * 3
    else:
        energy_charge = 100 * 2 + 100 * 3 + (units - 200) * 5

    fixed_charge = 50

    if units <= 100:
        rebate = 20
    else:
        rebate = 0

    total_bill = energy_charge + fixed_charge - rebate

    print("\n--- Electricity Bill ---")
    print("Units consumed:", units)
    print("Energy charge: Rs.", energy_charge)
    print("Fixed charge: Rs.", fixed_charge)
    print("Rebate: Rs.", rebate)
    print("Total bill: Rs.", total_bill)
```

```
--- Electricity Bill ---
Units consumed: 80
Energy charge: Rs. 160
Fixed charge: Rs. 50
Rebate: Rs. 20
Total bill: Rs. 190

========== RESTART: C:/Users/sirit/OneDriv
Enter electricity units consumed: 150

--- Electricity Bill ---
Units consumed: 150
Energy charge: Rs. 350
Fixed charge: Rs. 50
Rebate: Rs. 0
Total bill: Rs. 400
```